

# Relation Algebra

## §1. Notation

Traditionally Adjunction  $F \dashv G$  is defined as  $F(x) \leq y$  iff  $x \leq G(y)$ , and you derive laws like this:

$$\begin{aligned} F(x) \leq F(x) &\implies x \leq G(F(x)) \implies F(x) \leq F(G(F(x))) \\ G(y) \leq G(y) &\implies F(G(y)) \leq y \implies G(F(G(y))) \leq G(y) \end{aligned}$$

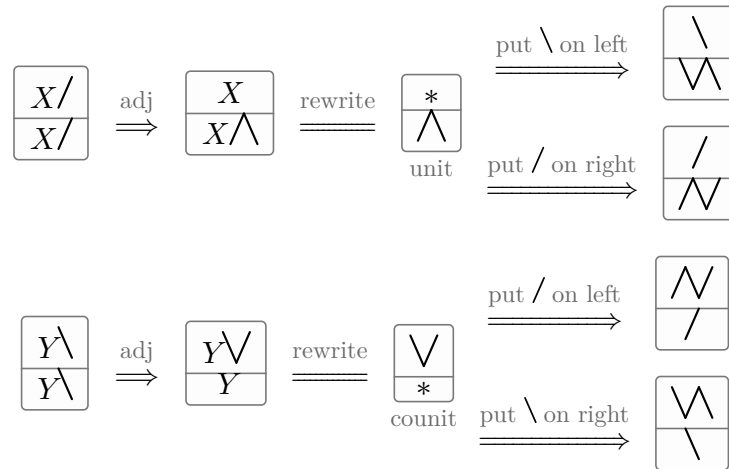
By define  $X : * \mapsto x$ ,  $Y : * \mapsto y$  and use diagram order, application can be replaced by composition:  $XF \leq Y$  iff  $X \leq YG$

(1a)

- write  $F$  as  $/$ ,  $G$  as  $\backslash$
- render  $a \leq b$  as  $\boxed{\frac{a}{b}}$
- **rewrite**  $X \leq XF$  as  $1 \leq F$

$$\boxed{\frac{X/}{Y}} = \boxed{\frac{X}{Y\backslash}}$$

adj



now we have  $\wedge = /$  and  $\vee = \backslash$ , and you just discovered string diagrams.

## §1.1. Adjunctions

(1.1a)

| $F \dashv G$                                           | $F$ 's type                                                           | monad $FG$                            | $1 \sqsubseteq FG$                                                                  | $GF \sqsubseteq 1$                                                                  | $F$ preserves $u$                                                 | $G$ preserves $n$                                                         | $FGF=F$                                                         | $GFG=G$                                                                       |
|--------------------------------------------------------|-----------------------------------------------------------------------|---------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------|---------------------------------------------------------------------------|-----------------------------------------------------------------|-------------------------------------------------------------------------------|
| $\triangleleft \dashv \triangleleft$                   | $A \rightarrow A \otimes A$                                           | $\triangleleft = 1$                   | $1 \sqsubseteq \triangleleft$                                                       | $\triangleleft \sqsubseteq 1$                                                       | $(R \cup S) \triangleleft = R \triangleleft \cup S \triangleleft$ | $(R \cap S) \triangleright = R \triangleright \cap S \triangleright$      | $\triangleleft \triangleleft = \triangleleft$                   | $\triangleright \triangleright = \triangleright$                              |
| $\multimap \dashv \multimap$                           | $A \rightarrow I$                                                     | $\multimap = T$                       | $1 \sqsubseteq \multimap$                                                           | $\multimap \sqsubseteq 1$                                                           | $(R \cup S) \multimap = R \multimap \cup S \multimap$             | $(R \cap S) \multimap = R \multimap \cap S \multimap$                     | $\multimap \multimap = \multimap$                               | $\multimap \multimap = \multimap$                                             |
| $\circ \dashv \circ$                                   | $(A \rightarrow B) \rightarrow (B \rightarrow A)$                     | $R^\circ \circ = R$                   | $R^\circ \circ = R$                                                                 | $R^\circ \circ = R$                                                                 | $(R \cup S)^\circ = S^\circ \cup R^\circ$                         | $(R \cap S)^\circ = R^\circ \cap S^\circ$                                 | $R^\circ \circ = R$                                             | $R^\circ \circ = R$                                                           |
| $\circ \dashv \triangleleft \dashv \multimap$          | $(X \otimes A \rightarrow Y) \rightarrow (X \rightarrow Y \otimes A)$ | $1$                                   | $(\circ \triangleleft \otimes 1) = 1$<br>$(1 \otimes \triangleright \multimap) = 1$ | $(1 \otimes \circ \triangleleft) = 1$<br>$(\triangleright \multimap \otimes 1) = 1$ | $=$                                                               | $=$                                                                       | $=$                                                             | $=$                                                                           |
| $\triangleright \dashv \multimap \dashv \triangleleft$ | $(X \rightarrow Y \otimes A) \rightarrow (X \otimes A \rightarrow Y)$ | $1$                                   | $(1 \otimes \circ \triangleleft) = 1$<br>$(\triangleright \multimap \otimes 1) = 1$ | $(\circ \triangleleft \otimes 1) = 1$<br>$(1 \otimes \triangleright \multimap) = 1$ | $=$                                                               | $=$                                                                       | $=$                                                             | $=$                                                                           |
| $\Delta \dashv \cap$                                   | $(A \rightarrow B) \rightarrow (A \rightarrow B)^2$                   | $R \cap R = R$                        | $R \cap R = R$                                                                      | $R \cap S \sqsubseteq R$                                                            | $\Delta(R \cup S) = \Delta R \cup \Delta S$                       | $(R \cap T) \cap (S \cap U) = (R \cap S) \cap (T \cap U)$                 | $R \cap R = R$                                                  | $R \cap R = R$                                                                |
| $u \dashv \Delta$                                      | $(A \rightarrow B)^2 \rightarrow (A \rightarrow B)$                   | $(R, S) \mapsto (R \cup S, R \cap S)$ | $R \sqsubseteq R \cup S$                                                            | $R \cup R = R$                                                                      | $(R \cup T) \cup (S \cup U) = (R \cup S) \cup (T \cup U)$         | $\Delta(R \cap S) = \Delta R \cap \Delta S$                               | $R \cup R = R$                                                  | $R \cup R = R$                                                                |
| $\emptyset \dashv !$                                   | $\{*\} \rightarrow (A \rightarrow B)$                                 |                                       |                                                                                     | $\emptyset \sqsubseteq R$                                                           |                                                                   |                                                                           |                                                                 |                                                                               |
| $\cdot S \dashv / S$                                   | $(A \rightarrow B) \rightarrow (A \rightarrow C)$                     | $S/S$                                 | $R \sqsubseteq (RS)/S$                                                              | $(T/S) \sqsubseteq T$                                                               | $(S \cup T)R = SR \cup TR$                                        | $(R \cap R')/S = (R/S) \cap (R'/S)$                                       | $((RS)/S)S = RS$                                                | $((T/S)S)/S = T/S$                                                            |
| $S \cdot \dashv S \setminus$                           | $(B \rightarrow C) \rightarrow (A \rightarrow C)$                     | $S \setminus S$                       | $R \sqsubseteq S \setminus (SR)$                                                    | $S(S \setminus T) \sqsubseteq T$                                                    | $R(S \cup T) = RS \cup RT$                                        | $S \setminus (R \cap R') = (S \setminus R) \cap (S \setminus R')$         | $S(S \setminus (SR)) = SR$                                      | $S \setminus (S(S \setminus T)) = S \setminus T$                              |
| $R \cap \dashv R \Rightarrow$                          | $(A \rightarrow B) \rightarrow (A \rightarrow B)$                     | $X \mapsto R \Rightarrow (X \cap R)$  | $X \sqsubseteq R \Rightarrow (X \cap R)$                                            | $R \cap (R \Rightarrow Y) \sqsubseteq Y$                                            | $R \cap (X \cup Y) = (R \cap X) \cup (R \cap Y)$                  | $R \Rightarrow S \cap T = (R \Rightarrow S) \cap (R \Rightarrow T)$       | $R \cap (R \Rightarrow (X \cap R)) = X \cap R$                  | $R \Rightarrow (R \cap (R \Rightarrow Y)) = R \Rightarrow Y$                  |
| $\mathcal{D} \dashv \cdot T$                           | $(A \rightarrow B) \rightarrow \text{Cor } A$                         | $R \mapsto (\mathcal{D}R)T$           | $R \sqsubseteq (\mathcal{D}R)T$                                                     | $\mathcal{D}(AT) \sqsubseteq A$                                                     | $\mathcal{D}(R \cup S) = \mathcal{D}R \cup \mathcal{D}S$          | $(A \cap B)T = AT \cap BT$                                                | $\mathcal{D}((\mathcal{D}R)T) = \mathcal{D}R$                   | $(\mathcal{D}(AT))T = AT$                                                     |
| $\mathcal{R} \dashv T \cdot$                           | $(A \rightarrow B) \rightarrow \text{Cor } B$                         | $R \mapsto T(\mathcal{R}R)$           | $R \sqsubseteq T(\mathcal{R}R)$                                                     | $\mathcal{R}(TA) \sqsubseteq A$                                                     | $\mathcal{R}(R \cup S) = \mathcal{R}R \cup \mathcal{R}S$          | $T(A \cap B) = TA \cap TB$                                                | $\mathcal{R}(T(\mathcal{R}R)) = \mathcal{R}R$                   | $T(\mathcal{R}(TA)) = TA$                                                     |
| $\cdot f \dashv \cdot f^\circ$                         | $(A \rightarrow B) \rightarrow (A \rightarrow C)$                     | $ff^\circ$                            | $\text{Map}(f) \Rightarrow 1 \sqsubseteq ff^\circ$                                  | $f^\circ f \sqsubseteq 1$                                                           | $(S \cup T)R = SR \cup TR$                                        | $\text{Simple}(g) \Rightarrow (X \cap Y)g^\circ = Xg^\circ \cap Yg^\circ$ | $ff^\circ f = f$                                                | $f^\circ ff^\circ = f^\circ$                                                  |
| $f^\circ \dashv \cdot f \cdot$                         | $(A \rightarrow C) \rightarrow (B \rightarrow C)$                     | $ff^\circ$                            | $\text{Map}(f) \Rightarrow 1 \sqsubseteq ff^\circ$                                  | $f^\circ f \sqsubseteq 1$                                                           | $R(S \cup T) = RS \cup RT$                                        | $f(X \cap Y) = fX \cap fY$                                                | $ff^\circ f = f$                                                | $f^\circ ff^\circ = f^\circ$                                                  |
| $i \dashv E$                                           | $\text{Map} \dashv \text{Rel}$                                        | $E$                                   | $\frac{1}{\exists} : A \rightarrow EA$                                              | $\exists : EB \rightarrow B$                                                        | —                                                                 | —                                                                         | $\frac{\exists}{\exists} = 1$                                   | $\frac{R}{\exists} \exists = R$                                               |
| $\cdot \exists \dashv \frac{\exists}{\exists}$         | $\text{Map}(A, EB) \rightarrow (A \rightarrow B)$                     | $1$                                   | $\frac{f\exists}{\exists} = f$                                                      | $\frac{R}{\exists} \exists = R$                                                     | —                                                                 | —                                                                         | $\frac{\frac{R}{\exists} \exists}{\exists} = \frac{R}{\exists}$ | $\frac{\frac{R}{\exists} \exists}{\exists} = \frac{R}{\exists}$               |
| $\frac{\exists}{\exists} \dashv \cdot \exists$         | $(A \rightarrow B) \rightarrow \text{Map}(A, EB)$                     | $1$                                   | $\frac{R}{\exists} \exists = R$                                                     | $\frac{f\exists}{\exists} = f$                                                      | —                                                                 | —                                                                         | $\frac{\frac{R}{\exists} \exists}{\exists} = \frac{R}{\exists}$ | $\frac{\frac{f\exists}{\exists} \exists}{\exists} = \frac{f\exists}{\exists}$ |

## §1.2. Composing adjunctions

(1.2a)

$$\begin{array}{c}
 \begin{array}{ccccc}
 \boxed{\begin{array}{c} X \\ f(R/T) \end{array}} & \xLeftrightarrow{f^\circ \cdot \dashv f \cdot} & \boxed{\begin{array}{c} f^\circ X \\ R/T \end{array}} & \xLeftrightarrow{\cdot T \dashv / T} & \boxed{\begin{array}{c} (f^\circ X)T \\ R \end{array}} \\
 \text{f a map} & & & & \\
 \xLeftrightarrow{\text{associativity}} & \boxed{\begin{array}{c} f^\circ(XT) \\ R \end{array}} & \xLeftrightarrow{f^\circ \cdot \dashv f \cdot} & \boxed{\begin{array}{c} XT \\ fR \end{array}} & \xLeftrightarrow{\cdot T \dashv / T} & \boxed{\begin{array}{c} X \\ (fR)/T \end{array}} \\
 \xRightarrow{\text{indirect equality}} & & & & \boxed{\begin{array}{c} f(R/T) \\ (fR)/T \end{array}}
 \end{array}$$

$$\text{Map}(f) \Rightarrow f(R/S) = fR/S$$

(1.2b)

|                            | $\cdot T$                                    | $T \cdot$                                                 | $\cdot g$                                         | $g^\circ \cdot$                              | $Tn$                                                      | $^\circ$                                                                                                                            | $\leftarrow \triangleleft$                                                                                                          | $\Delta$                                                       |
|----------------------------|----------------------------------------------|-----------------------------------------------------------|---------------------------------------------------|----------------------------------------------|-----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| $\cdot S$                  | $R/S_1 \ S_2=R/S_2/S_1$                      | $S \setminus (R/W) = (S \setminus R)/W$                   | $R/(Sg) = (Rg^\circ)/S$                           | $\text{Map}(f) \implies f(R/S) = fR/S$       | —                                                         | $(Y/S)^\circ = S^\circ \setminus (Y^\circ)$                                                                                         | $(RS)^\wedge = R^\wedge (S \otimes 1)$                                                                                              | $(RnR')/S = (R/S) \cap (R'/S)$                                 |
| $S \cdot$                  | $S \setminus (R/W) = (S \setminus R)/W$      | $S_1 \ S_2 \setminus R = S_2 \setminus (S_1 \setminus R)$ | $S \setminus (Rg^\circ) = (S \setminus R)g^\circ$ | $(g^\circ S) \setminus R = S \setminus (gR)$ | —                                                         | $(S \setminus Y)^\circ = Y^\circ / S^\circ$                                                                                         | $((S \otimes 1)R)^\wedge = SR^\wedge$                                                                                               | $S \setminus (RnR') = (S \setminus R) \cap (S \setminus R')$   |
| $\cdot f$                  | $\text{Map}(f) \implies R/fS = (R/S)f^\circ$ | $T \setminus (Rf^\circ) = (T \setminus R)f^\circ$         | $(RS)^\circ = S^\circ R^\circ$                    |                                              |                                                           | $(RS)^\circ = S^\circ R^\circ$                                                                                                      |                                                                                                                                     | $\text{Simple}(g) \implies (XnY)g^\circ = Xg^\circ nYg^\circ$  |
| $f^\circ \cdot$            | $\text{Map}(f) \implies f(R/S) = fR/S$       | $(Tf^\circ) \setminus R = f(T \setminus R)$               | —                                                 | $(RS)^\circ = S^\circ R^\circ$               |                                                           | $(RS)^\circ = S^\circ R^\circ$                                                                                                      |                                                                                                                                     | $f(T_1 n T_2) = fT_1 n fT_2$                                   |
| $Rn$                       | —                                            | —                                                         | —                                                 | —                                            | $R \Rightarrow S \Rightarrow T = RnS \Rightarrow T$       | $(R \Rightarrow Y)^\circ = R^\circ \Rightarrow (Y^\circ)$                                                                           | —                                                                                                                                   | $R \Rightarrow SnT = (R \Rightarrow S) \cap (R \Rightarrow T)$ |
| $^\circ$                   | $(Y/T)^\circ = T^\circ \setminus (Y^\circ)$  | $(T \setminus Y)^\circ = Y^\circ / T^\circ$               | $(RS)^\circ = S^\circ R^\circ$                    | $(RS)^\circ = S^\circ R^\circ$               | $(T \Rightarrow Y)^\circ = T^\circ \Rightarrow (Y^\circ)$ | $R^\circ \circ = R$                                                                                                                 | $(R^\wedge)^\circ = (R^\circ \otimes 1)$<br>$(1 \otimes \triangleright \dashv \circ)$<br>both bends: the $^\circ$ section's display | $(T_1 n T_2)^\circ = T_1^\circ n T_2^\circ$                    |
| $\leftarrow \triangleleft$ | $(RT)^\wedge = R^\wedge (T \otimes 1)$       | $((T \otimes 1)R)^\wedge = TR^\wedge$                     | —                                                 | —                                            | —                                                         | $(R^\wedge)^\circ = (R^\circ \otimes 1)$<br>$(1 \otimes \triangleright \dashv \circ)$<br>both bends: the $^\circ$ section's display | —                                                                                                                                   | —                                                              |
| $u$                        | $(S \cup T)R = SR \cup TR$                   | $R(S \cup T) = RS \cup RT$                                | $(S \cup T)R = SR \cup TR$                        | $R(S \cup T) = RS \cup RT$                   | $Rn(S \cup T) = RnS \cup RnT$                             | $(R \cup S)^\circ = S^\circ \cup R^\circ$                                                                                           |                                                                                                                                     |                                                                |
| $\emptyset$                | $\emptyset R = \emptyset$                    | $R \emptyset = \emptyset$                                 | $\emptyset R = \emptyset$                         | $R \emptyset = \emptyset$                    | $Rn \emptyset = \emptyset$                                | $\emptyset^\circ = \emptyset$                                                                                                       |                                                                                                                                     |                                                                |

### §1.3. Adjoint triples

Beyond  $u \dashv \Delta \dashv n$ , which §1.1 already carries as two rows, the table holds one adjoint triple with content:  $\exists_f \dashv f^* \dashv \forall_f$  along a map  $f$ , once on each side of the composite.  $f : A \rightarrow B$  throughout this subsection.

(1.3a)

$$\begin{aligned} \cdot f \dashv \cdot f^\circ \dashv /f^\circ \\ f^\circ \cdot \dashv f \cdot \dashv f \setminus \end{aligned}$$

The first two links of each chain are the map shunting rules, and the third is §1.1's division row read at a chosen divisor:  $\cdot S \dashv /S$  at  $S := f^\circ$ , and  $S \cdot \dashv S \setminus$  at  $S := f$ .

The third link is not the first over again. For a map  $f$ ,  $/f$  collapses to  $\cdot f^\circ$ ; being a map is exactly what that collapse spends  $\text{Map}(f) \implies R/fS = (R/S)f^\circ$  —  $f^\circ$  is not a map, so  $/f^\circ$  does not collapse in turn, and the three operators stay distinct. On  $\text{Rel}(\mathbb{I}, A) = \text{EA}$  they are the image triple:

(1.3b)

| operator        | acts by                                   | name                      |
|-----------------|-------------------------------------------|---------------------------|
| $\cdot f$       | $S \mapsto f[S]$                          | direct image, $\exists_f$ |
| $\cdot f^\circ$ | $T \mapsto f^{-1}[T]$                     | inverse image, $f^*$      |
| $/f^\circ$      | $S \mapsto \{b : f^{-1}(b) \subseteq S\}$ | $\forall_f$               |

§1.1's  $\mathcal{D} \dashv \cdot \tau$  is this same chain along the projection  $A \otimes B \rightarrow A$ , read through  $\text{Rel}(A, B) = E(A \otimes B)$ :  $\mathcal{D}R = \{(a, a) : \exists b. aRb\}$ ,  $AT$  is the relation that ignores  $b$  altogether, and the third link is  $R \mapsto 1 \cap R / \tau = \{(a, a) : \forall b. aRb\}$ .

(1.3c)

$$\mathcal{D} \dashv \cdot \tau \dashv 1 \cap \cdot / \tau$$

Three is where it stops, and one map breaks both ends. Take  $A = \{a_1, a_2\}$  and  $B = \{b\}$ :  $f[\{a_1\}] \cap f[\{a_2\}] = \{b\}$  while  $f[\{a_1\} \cap \{a_2\}] = \emptyset$ , so  $\cdot f$  does not preserve meets and has no left adjoint, and the same two subsets give  $\forall_f(\{a_1\} \cup \{a_2\}) = \{b\}$  against  $\forall_f\{a_1\} \cup \forall_f\{a_2\} = \emptyset$ , so  $/f^\circ$  does not preserve joins and has no right adjoint. A monic  $f$  restores the binary case and no more —  $\tau f$  still reaches only  $\text{im } f$ , and  $\forall_f f \emptyset = B \setminus \text{im } f$  is still not  $\emptyset$ . The chain extends only when  $f^\circ$  is a map as well, and then  $/f^\circ = \cdot f^\circ = \cdot f$  and it repeats forever. For an  $S$  with neither  $S$  nor  $S^\circ$  a map there is no triple at all:  $\cdot S \dashv /S$  is two links and stops.

(1.2b)

The rows that chain forever say nothing by it:  $\circ$  is an order-isomorphism of hom-posets, so it is adjoint to itself on both sides and  $\circ \dashv \circ \dashv \circ$  has no end, and §1.1's other row of that kind,  $\circ \triangleleft \dashv \triangleright \circ$ , goes the same way.

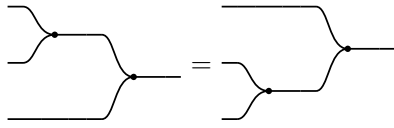
## §2. Relations

(2a)

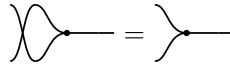
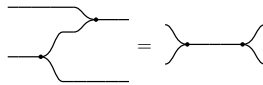
$\mathbf{Rel}$  is a poset-enriched category with  $(\otimes, \circ)$  where  $\otimes$  is commutative cartesian product, and converse  $\circ^{-1}$ , and

$C: (\triangleright : A \otimes A \rightarrow A, \circ : I \rightarrow A)$  a commutative monoid with  $C^\circ \dashv C$ . We write  $C^\circ$  as  $(\triangleleft : A \rightarrow A \otimes A, \circ : A \rightarrow I)$

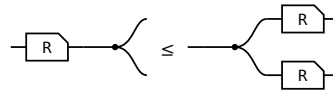
(2b)



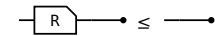
$\triangleright$  associative



$\triangleright$  commutative



$\circ$  is its unit



Frobenius, one half — the other is its  $\circ$

$R \triangleleft \leq \triangleleft (R \otimes R)$

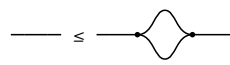
$R \circ \leq \circ$

### §2.1. $\forall$ object $A$ , $(A, \triangleleft, \circ) \dashv (A, \triangleright, \circ)$

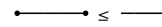
(2.1a)



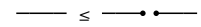
$\triangleright \triangleleft \leq 1$



$1 \leq \triangleleft$



$\circ \leq 1$



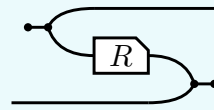
$1 \leq \circ$

The last of these is the only one that makes a picture **bigger**, and it is worth a name: **a wire is below the cut wire**. In  $\mathbf{Rel}$  it reads  $\{(a, a)\} \subseteq A \times A$  — cut a wire and its two ends stop having to agree, so cutting can only add pairs. It is the one weakening this calculus gives away for free.

### §3. $\circ : \mathcal{C}^{\text{op}} \rightarrow \mathcal{C}$ is a 2 functor

(3a)

$\circ$  is primitive, part of the data the first section lists. It turns both of  $R$ 's wires round, and the Frobenius structure DRAWS that — the picture and the formula below are  $R^\circ$ , not its definition:



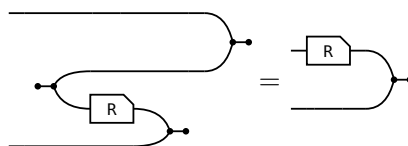
$$R^\circ = (\circ \triangleleft \otimes 1) (1 \otimes R \otimes 1) (1 \otimes \triangleright \multimap \circ)$$

where  $\circ \triangleleft : \mathbb{I} \rightarrow \mathbf{a} \otimes \mathbf{a}$  opens a pair of wires out of nothing and  $\triangleright \circ : \mathbf{b} \otimes \mathbf{b} \rightarrow \mathbb{I}$  closes one, so the input of  $R^\circ$  is where the output of  $R$  was. Taking that formula as the definition would now be circular:  $\triangleleft$  and  $\multimap$  are  $\triangleright^\circ$  and  $\circ^\circ$ . The laws of  $\circ$  are that it is a contravariant 2-functor  $\circ : \mathcal{C}^{\text{op}} \rightarrow \mathcal{C}$ :

$$(i) \ 1^\circ = 1 \quad (ii) \ (RS)^\circ = S^\circ R^\circ \quad (iii) \ (R \otimes S)^\circ = R^\circ \otimes S^\circ \quad (iv) \ R \leq S \text{ implies } R^\circ \leq S^\circ$$

#### §3.1. The slide

The one rule (ii) and (iii) use, and each of them uses it twice. A converse facing a merge on the lower strand is the box itself, upright, on the upper one:



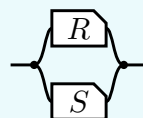
(3.1a)

$R^\circ$  is DEFINED as the bending of  $(R \otimes 1) \triangleright \circ$ , so the slide claims only that unbending it gives that back — and unbending undoes bending for every arrow. That is the snake above with a passenger: the  $\circ \triangleleft$  bends the  $\mathbf{a}$  strand down and the  $\triangleright \circ$  brings it back up, while the  $\mathbf{b}$  strand rides through untouched. Nothing else is spent below.

## §4. $\cap$ is a commutative idempotent monoid on every hom-set

(4a)

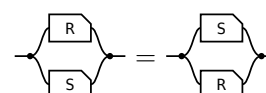
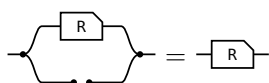
The **meet** of  $R, S : a \rightarrow b$ , the paper's **convolution**, is  $R \cap S := \langle \downarrow (R \otimes S) \rangle$  — copy the input, run  $R$  and  $S$  on the two copies, merge the results — so what comes out is what both of them do.



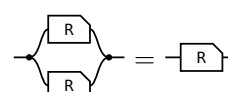
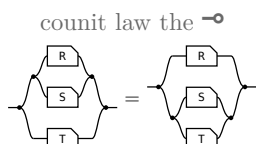
transcribed: a definition has no statement to export

On every hom-set it is associative, commutative and idempotent, with unit the maximal arrow  $\top = \text{copy} \circ \text{merge}$  (the paper's Lemma 4.11).

(4b)



**unit:** one half of  $\top$  per end — the merge's unit law absorbs the  $\text{copy}$ , the copy's counit law the  $\text{merge}$  **commutative:**  $\sigma$  crosses  $R \otimes S$  by naturality and is absorbed by cocommutativity and commutativity



**associative:** coassociativity and associativity;  $\otimes$  re-brackets for nothing, **idempotent:** the one that is not bookkeeping — the lax copy law is the whole of it, worked in allegory2

So  $\leq$  is the order this monoid induces.  $R \cap S \leq R$  comes from the unit, and idempotency turns anything under both  $S$  and  $T$  into something under  $S \cap T$ , since  $R = R \cap R \leq S \cap T$ .

And one law relating  $\cap$  to composition, which is **not** an equation:

(4c)

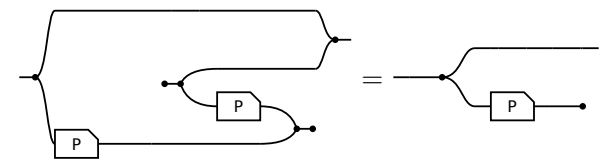
| semi-distributivity, and what supplies it                                                                                                                         | picture |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| $R \cap (S \cap T) \leq R \cap S \cap R \cap T$ — the lax copy law. Equality exactly when $R$ is single valued: the Maps section's $F(R \cap S) = F R \cap F S$ . |         |

## §5. Domain and range

(5a)

The **domain**  $\text{dom}(R) \triangleq 1 \cap R R^\circ$  and the **range**  $\text{ran}(R) \triangleq \text{dom}(R^\circ)$ .

(5b)



$\triangleright$  lands the return leg back on the value  $\triangleleft$  handed out,  
so  $R^\circ \triangleright$  cuts to  $\circ$  — Frobenius

Running  $R$  and throwing the result away leaves only the fact that  $R$  could fire, and  $\text{ran}(R)$  is the same picture with the box mirrored. In  $\text{Rel}$  both steps are  $\{(a, a) : \exists b. a R b\}$ .

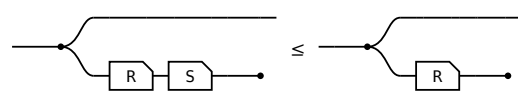
(5c)

|                                                                                          |
|------------------------------------------------------------------------------------------|
| $\text{dom}(R) \triangleq 1 \cap R R^\circ$                                              |
| $\text{Coreflexive}(X) \implies \text{dom}(R) \subseteq X \iff R \subseteq X R$          |
| $\text{dom}(RS) \subseteq \text{dom}(R)$                                                 |
| $\text{dom}(R \cap S) = 1 \cap S R^\circ$                                                |
| $R \text{ entire} \iff \text{Entire}(R) \triangleq \text{dom}(R) = 1$                    |
| $R \text{ simple} \iff \text{Simple}(R) \triangleq R^\circ R \subseteq 1$                |
| $R \text{ a map} \iff \text{Map}(R) \triangleq \text{Entire}(R) \wedge \text{Simple}(R)$ |
| $\text{Entire}(R) \wedge \text{Entire}(S) \implies \text{Entire}(RS)$                    |
| $\text{Simple}(f) \wedge \text{Simple}(g) \implies \text{Simple}(fg)$                    |
| $\text{Map}(f) \wedge \text{Map}(g) \implies \text{Map}(fg)$                             |
| $\text{Entire}(RS) \implies \text{Entire}(R)$                                            |

### §5.1. Sliding the discard

$\text{dom}(RS) \subseteq \text{dom}(R)$ , and a single glyph for **dom** would have nothing to slide: with the box and the discard drawn apart, the law is one dot walking back along the lower strand.

(5.1a)



$S \circ \circ \subseteq \circ$ , the lax axiom for  $\circ$  in the first section  
— the discard slides back past  $S$

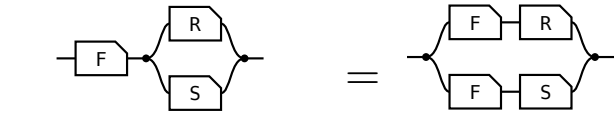
Equality is  $S$  entire, which is the same picture read as  $\text{dom}(R) = 1 \iff R \text{ entire}$ .

## §6. Maps

(6a)

|                                                                     |                                                 |
|---------------------------------------------------------------------|-------------------------------------------------|
| <b>surjective</b><br>$1 \in R^\circ R$ , that is, $R^\circ$ entire. | <b>single valued</b><br>$R^\circ R \subseteq 1$ |
| <b>entire</b><br>$1 \in RR^\circ$ . With single valued, a map.      | <b>injective</b><br>$RR^\circ \subseteq 1$      |

(6b)

| law                                                                            | picture                                                                                                                               |
|--------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| $\text{Simple}(F) \Rightarrow F(R \cap S) = F R \cap F S$<br>$F$ single valued | <br>one person who admires both      a at A, b at B |

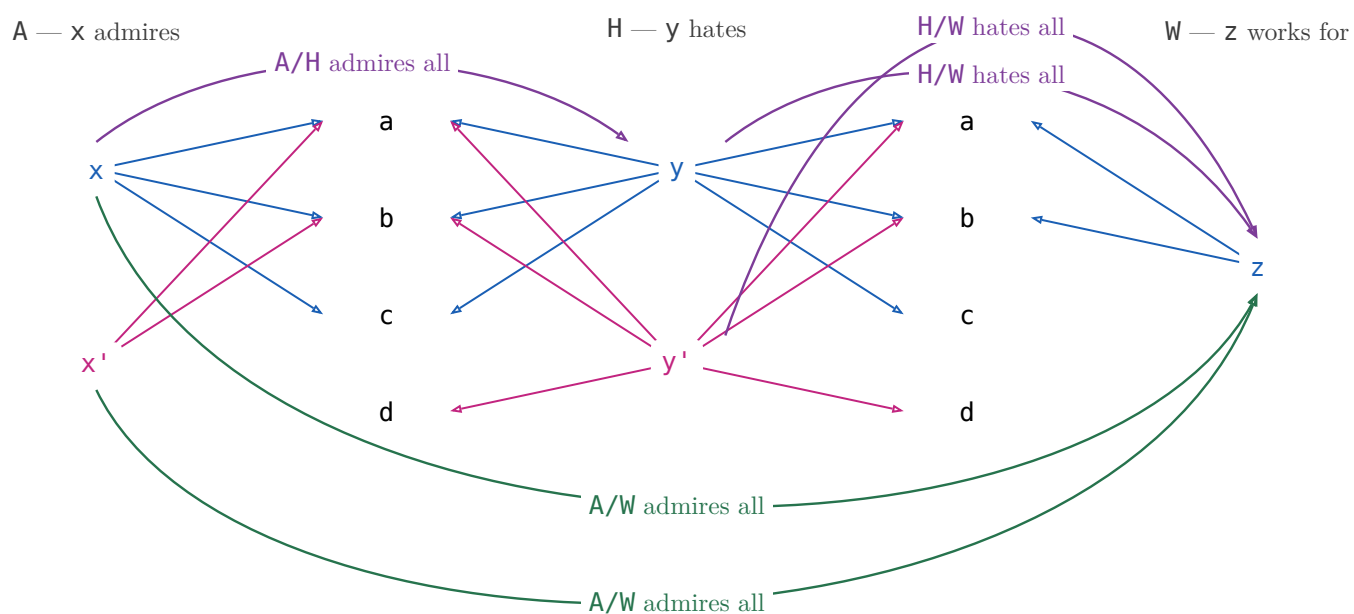
## §8. / is all of

(8a)

$x (R/S) y \Leftrightarrow \forall p. y S p \rightarrow x R p$   
 $x (S \setminus R) y \Leftrightarrow \forall p. p S x \rightarrow p R y$   
/compares images:  $S(y) \subseteq R(x)$ . \ compares preimages:  $S^\circ(x) \subseteq R^\circ(y)$ .  
example: A admires, H hates, W works for  
 $x (A/H) y$  — x admires everyone y hates.  
 $x (H \setminus A) y$  — everyone who hates x admires y.

§8.1.  $(R/S)(S/W) \subseteq R/W$ 

(8.1a)



(8.1b)

$x (A/H) y$  — x admires everyone y hates  
 $y (H/W) z$  — y hates everyone z works for  
 $x (A/W) z$  — x admires everyone z works for

$(A/H)(H/W)$  is a path:  $x \rightarrow y \rightarrow z$ , and that is all of it.  $A/W$  also holds of  $x'$ , who admires everyone z works for — but  $x'$  does not admire everyone anybody hates, so nothing composes to it. The missing path is exactly the strictness of  $(R/S)(S/W) \subseteq R/W$ .

|                                                                                                                                                                                                                 |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| $X \in R/S \iff XS \in R$<br>$X$ is any $x$ -to- $y$ pairing; one that only pairs $x$ with a $y$ such that $x$ admires everyone $y$ hates lies inside $R/S$ , and $R/S$ is the largest such.                    |  |
| $X \in S \setminus R \iff SX \in R$<br>The mirror — divide on the left when $x$ comes first.                                                                                                                    |  |
| $(R/S)S \in R$<br>There is a $y$ such that $x$ admires everyone $y$ hates, and $p$ is one of the people $y$ hates — then $x$ admires $p$ too. Strict at $S=\emptyset$ : $R/S$ is everyone, $(R/S)S=\emptyset$ . |  |
| $S \setminus (S \setminus R) \in R$<br>The mirror.                                                                                                                                                              |  |
| <b>associate:</b> $R/(S_1 S_2) = (R/S_2)/S_1$<br>A friend's enemy is two hops: divide by the far end first.                                                                                                     |  |
| $(S_1 S_2) \setminus R = S_2 \setminus (S_1 \setminus R)$<br>The mirror.                                                                                                                                        |  |
| <b>maps:</b> $f \ (R/S) = (fR)/S$<br>Rename $x$ before or after dividing — the licence to write $fR/S$ .                                                                                                        |  |
| $R/(fS) = (R/S) f^\circ$<br>Rename $y$ : a map leaves a denominator as $f^\circ$ outside the box.                                                                                                               |  |
| $(R/S)(S/W) \in R/W$<br>Someone who admires all of a hate-set that already covers everyone $z$ works for admires those people too.                                                                              |  |
| $1 \in R/R$<br>$R/R$ runs admirer to admirer: each admires everyone they admire. Strict: two people who each admire only $a$ and $b$ admire each other's idols too, and still stay two people.                  |  |
| $(R/R)(R/R) = R/R$<br>$R/R$ is the preorder <b>admires at least as much as</b> , and a preorder is idempotent. Freyd writes $\sqsubseteq$ ; with $1 \in R/R$ above it is an equality.                           |  |
| $(R/R)R = R$<br>Reaching $p$ through someone whose idols $x$ fully admires is reaching $p$ directly, since $x$ admires their own idols.                                                                         |  |
| $R/1 = R$<br>Dividing by $1$ : $p$ 's set is just $\{p\}$ , so admiring all of it is admiring $p$ .                                                                                                             |  |
| $R/(S_1 \cup S_2) = R/S_1 \cap R/S_2$<br>Admiring a combined hate-set is admiring each set in full.                                                                                                             |  |
| $S \setminus (R/W) = (S \setminus R)/W$<br>Which is why $S \setminus R/W$ needs no bracket.                                                                                                                     |  |

Fifteen laws, fifteen pictures, and not one shows a generator:  $\cap$ ,  $\cup$ ,  $\circ$  and composition are what the Frobenius generators build, and  $/$  is none of those — it is posited, with nothing to unfold.

# §9. $\frac{R}{S}$

(9a)

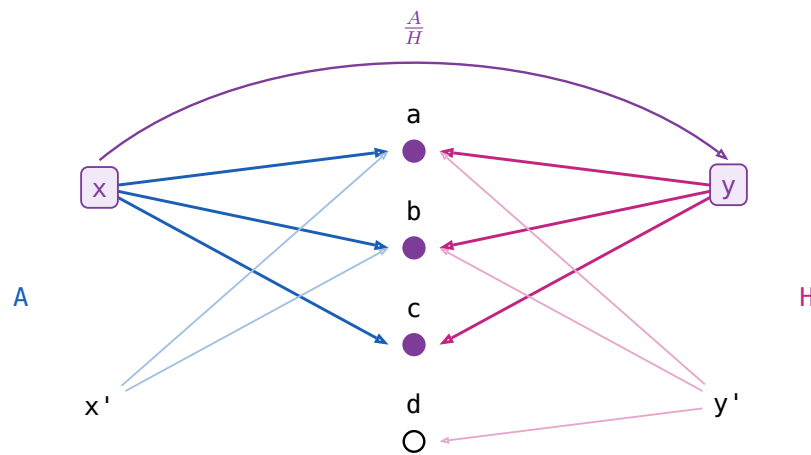
$R/\frac{S}{S} \triangleq (R/S) \cap (S/R)^\circ$  what  $R$  sends  $x$  to and nothing else is what  $S$  sends  $y$  to. In Rel  $x$  and  $y$  has the same image:  $\forall p. (x R p \iff y S p)$

(9b)

$x (A/H) y$  —  $x$  admires everyone  $y$  hates  
 $x ((H/A)^\circ) y$  —  $x$  admires only people  $y$  hates  
 $x ((A/H) \cap (H/A)^\circ) y$  —  $x$  admires only and all whom  $y$  hates

$x$  admires exactly whom  $y$  hates:  $/$  is **all of**,  $\frac{R}{S}$  is **only all of**.

(9c)



(9d)

| law                                                                                                                                                                               | the reading                                                                                                                                              |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| $X \in R/\frac{S}{S} \iff X S \in R \wedge X^\circ R \in S$ $X$ sits inside the matching exactly when it carries $S$ back inside $R$ and, reversed, carries $R$ back inside $S$ . | $X$ may pair $x$ with $y$ only when $x$ admires exactly whom $y$ hates. Both halves must typecheck, so the operation is <b>partial</b> .                 |
| $(R/\frac{S}{S})^\circ = S/\frac{R}{R}$ reversing the matching swaps the two relations.                                                                                           | Matching is symmetric.                                                                                                                                   |
| $(R/\frac{S}{S})(S/\frac{W}{W}) \in R/\frac{W}{W}$ a match from $R$ to $S$ followed by one from $S$ to $W$ is a match from $R$ to $W$ .                                           | And transitive.                                                                                                                                          |
| $(R/\frac{S}{S}) S \in R$ following the matching by $S$ lands inside $R$ .                                                                                                        | $(\exists y. x(\frac{R}{S})y \wedge y S p) \rightarrow x R p$<br>$x$ <b>only</b> admires whom $y$ hates<br>$\frac{R}{S} S = \text{Dom}(\frac{R}{S}) R$   |
| $(R/\frac{S}{S}) R = R$ matching $R$ against itself and then following $R$ gives $R$ back.                                                                                        | $(\exists y. x(\frac{R}{R})y \wedge y R p) \iff x R p$<br>$x$ and $y$ <b>admire the same people</b><br>$y=x$ always qualifies ( $1 \in R \circ R$ below) |
| $1 \in R/\frac{S}{S} R$ every $x$ matches itself.                                                                                                                                 | $x(\frac{R}{R})y$ if $x$ and $y$ admires the same peoples.                                                                                               |
| $(R/\frac{S}{S})(R/\frac{S}{S}) = R/\frac{S}{S}$ matching twice matches no more pairs than matching once.                                                                         | So the relation <b>admires the same people</b> is an equivalence relation.                                                                               |
| $\text{Symmetric}(X) \implies X \in R/\frac{S}{S} R \iff X R \in R$ a symmetric $X$ sits inside the matching exactly when following it by $R$ adds nothing to $R$ .               | The largest symmetric arrow that leaves $R$ alone.                                                                                                       |
| $\text{simplePart}(R) \triangleq R/\frac{S}{S} 1$ matching $R$ against the identity keeps the $x$ whose image is a single point.                                                  | The people who admire exactly one person and nobody else. It equals $R$ only when $R$ is simple, unlike $R/1 = R$ .                                      |
| $\text{dom}(R/\frac{S}{S}) = 1 \cap (R/S)(S/R)$ the $x$ that match something are those the two one-sided divisions take round the loop back to $x$ .                              | Its domain is the <b>domain of simplicity</b> of $R$ .                                                                                                   |

## §10. Freyd's Power allegories

(10a)

A **power allegory** is a division allegory with one unary operation on arrows,  $\exists$  **epsilon**, subject to

$$\begin{aligned} \exists_R \square &= R\square, \quad \exists_R = \exists_R \square \\ 1 &\subseteq (R/\exists_R)(\exists_R/R) & \exists_R \text{ is } \mathbf{thick} \\ \exists_R &= 1 & \exists_R \text{ is } \mathbf{straight} \end{aligned}$$

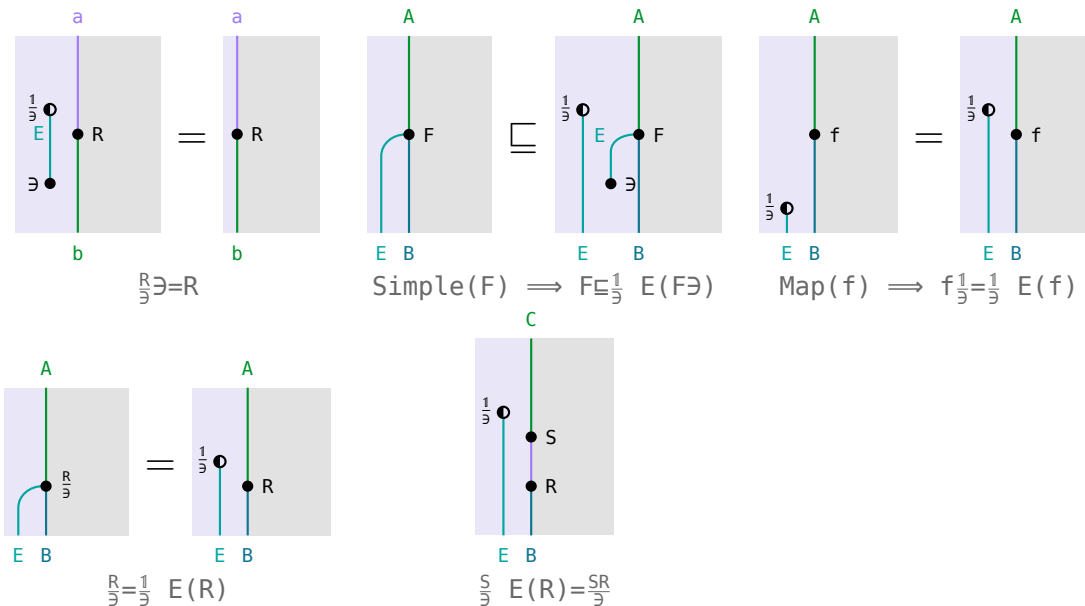
$R\square$  is  $R$ 's target, an identity arrow. For  $R : A \rightarrow B$  write  $\exists : EB \rightarrow B$ , dropping the subscript.

$$\in \triangleq \exists^\circ : A \rightarrow EA$$

(10b)

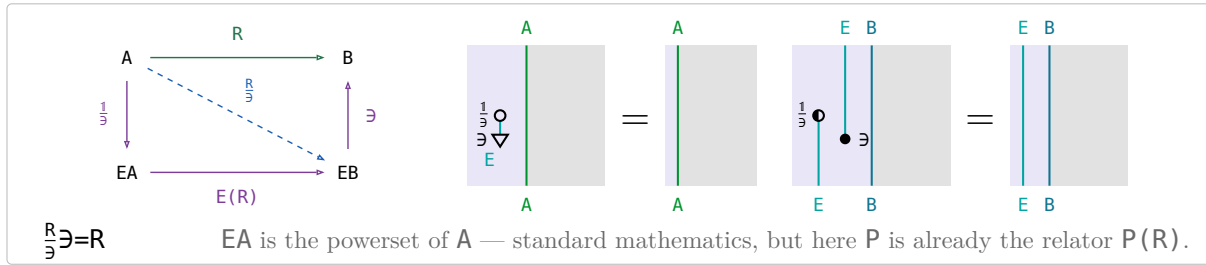
| law                                                                                                                                                             | the reading                                                                                                                                                                                      |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 $\exists_R \square = R\square, \quad \exists_R = \exists_R \square$                                                                                           | One $\exists$ per object, not per arrow.                                                                                                                                                         |
| 2 $\exists$ is <b>thick</b>                                                                                                                                     | <b>Comprehension:</b> every $x$ has a set of exactly the people $x$ admires.                                                                                                                     |
| 3 $\frac{R}{\exists} : A \rightarrow EB$ , for $R : A \rightarrow B$                                                                                            | convert a relation to a function. $a \frac{R}{\exists} = \{b \mid a R b\}$                                                                                                                       |
| 4 $\text{Map}(\frac{R}{\exists})$                                                                                                                               |                                                                                                                                                                                                  |
| 5 $\frac{R}{\exists} \exists = R$                                                                                                                               | reading the chosen set back through $\exists$ returns the relation.                                                                                                                              |
| 6 $\text{Simple}(F) \Rightarrow F \subseteq \frac{F\exists}{\exists}$                                                                                           | A partial choice of sets is inside the total one.                                                                                                                                                |
| 7 the <b>singleton map</b> is monic: $\frac{1}{\exists}$<br>$(\frac{1}{\exists})^\circ \subseteq 1$ two points with the same one-person set are the same point. | The one-person set.                                                                                                                                                                              |
| 8 $\frac{\exists}{\exists} = 1$                                                                                                                                 | Make the set of a set, then read it back one level down.                                                                                                                                         |
| 9 <b>fusion:</b> $\text{Map}(f) \Rightarrow \frac{fR}{\exists} = f \frac{R}{\exists}$                                                                           | Naturality of the unit, $\text{Map}(f) \Rightarrow f \frac{1}{\exists} = \frac{1}{\exists} E(f)$ . renaming a point and then taking its one-person set is taking the set and renaming inside it. |
| 10 $\text{Map}(f) \wedge 1\square = \exists\square \Rightarrow \frac{f}{\exists} = f \frac{1}{\exists}$                                                         | Rename first or take singletons first — the fusion row above at $R=1$ .                                                                                                                          |
| 11 $R\square = \exists\square \Rightarrow R/\circ S = \frac{R}{\exists} (\frac{S}{\exists})^\circ$                                                              | $x$ and $y$ match when $R$ sends $x$ and $S$ sends $y$ to the same set.                                                                                                                          |
| 12 $E(R) \triangleq \frac{\exists R}{\exists}$                                                                                                                  | $E(R) : EA \rightarrow EB$ , image of a set of $A$                                                                                                                                               |
| 13 $\frac{R}{\exists} = \frac{1}{\exists} E(R)$                                                                                                                 | $\frac{1}{\exists} : x \mapsto \{x\}$                                                                                                                                                            |
| 14 $\frac{S}{\exists} E(R) = \frac{SR}{\exists}$                                                                                                                | absorption — the monad's composition law, $\frac{S}{\exists} \diamond \frac{R}{\exists} = \frac{SR}{\exists}$ , §10.3                                                                            |
| 15 $\exists \triangleq \exists/\exists$                                                                                                                         | $xs \supseteq ys \iff \forall a. ys \exists a \rightarrow xs \exists a$                                                                                                                          |
| 16 $\subseteq \triangleq \exists^\circ$                                                                                                                         | $xs \subseteq ys \iff ys \supseteq xs$                                                                                                                                                           |

(10c)



## §10.1. $\mathbf{i} \dashv \mathbf{E}$ Power Allegory defined as adjunction

(10.1a)



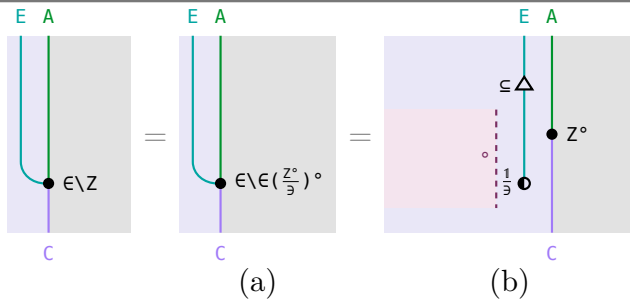
$\mathbf{i}$  is the inclusion  $\mathbf{Map}(\mathcal{A}) \rightarrow \mathcal{A}$ , doing nothing, so  $\mathbf{E}$  is both the functor and the monad  $\mathbf{i} \mathbf{E}$ .

## §10.2. $\mathbf{E} \setminus$ as a composite

$$\mathbf{E} \setminus \mathbf{Z} = \mathbf{E} \left( \frac{\mathbf{Z}^\circ}{\exists} \right)^\circ$$

(10.2a)

$\mathbf{x} \mathbf{s}$  is related to  $\mathbf{c}$  by  $\mathbf{E} \setminus \mathbf{Z}$  exactly when  $\mathbf{x} \mathbf{s} \subseteq \Lambda(\mathbf{Z}^\circ)(\mathbf{c})$ , the set of every  $\mathbf{a}$  with  $\mathbf{a} \mathbf{Z} \mathbf{c}$

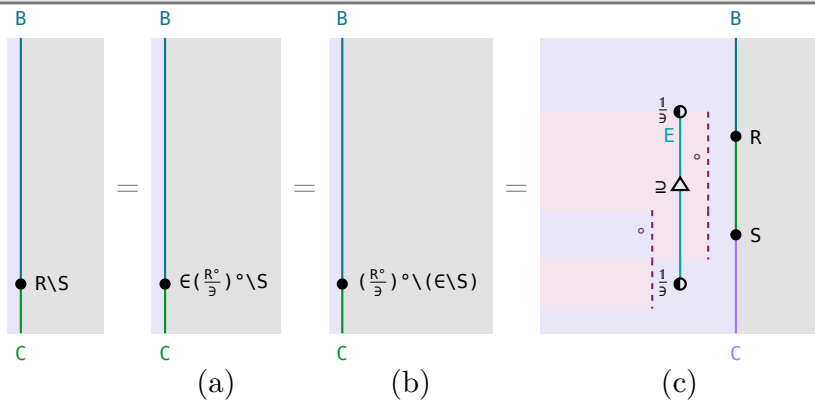


(b)  $X \setminus (Y f^\circ) = (X \setminus Y) f^\circ$ ,  $f$  a map

$$\mathbf{R} \setminus \mathbf{S} = \frac{\mathbf{R}^\circ}{\exists} \subseteq \left( \frac{\mathbf{S}^\circ}{\exists} \right)^\circ$$

(10.2b)

$\mathbf{b}$  is related to  $\mathbf{c}$  by  $\mathbf{R} \setminus \mathbf{S}$  exactly when  $\Lambda(\mathbf{R}^\circ)(\mathbf{b}) \subseteq \Lambda(\mathbf{S}^\circ)(\mathbf{c})$ , the  $\mathbf{R}$ -preimage of  $\mathbf{b}$  inside the  $\mathbf{S}$ -preimage of  $\mathbf{c}$



(b)  $(XY) \setminus S = Y \setminus (X \setminus S)$

(c)  $f^\circ \setminus X = f X$ ;  $E \setminus S = \subseteq \Lambda(S^\circ)^\circ$  — (10.2a)

### §10.3. $\mathcal{A} \cong \mathbf{Kleisli}(E)$

(10.3a)

$E(R) \triangleq \frac{\exists R}{\exists}$ ,  $\frac{1}{\exists} : A \rightarrow EA$ ,  $\text{union} = E(\exists) : E(EA) \rightarrow EA$

$f \diamond g \triangleq f E(g) \text{union}$ , for  $f : A \rightarrow EB$  and  $g : B \rightarrow EC$

the monad is on  $\mathbf{Map}(\mathcal{A})$ , not on the allegory:  $E$  is a relator on all relations, but  $\frac{1}{\exists}$ ,  $\text{union}$  and  $f E(g) \text{union}$  are maps, and the Kleisli construction happens where they live.

(10.3b)

$$\begin{array}{c}
 \boxed{\frac{S}{\exists} \diamond \frac{R}{\exists}} \xrightarrow{\text{definition of } \diamond, \text{union} = E(\exists)} \boxed{\frac{S}{\exists} E\left(\frac{R}{\exists}\right) E(\exists)} \xrightarrow{E \text{ a functor, } E(X)E(Y) = E(XY)} \boxed{\frac{S}{\exists} E\left(\frac{R}{\exists}\exists\right)} \\
 \xrightarrow{(10b), \frac{R}{\exists}\exists = R} \boxed{\frac{S}{\exists} E(R)} \xrightarrow{(10b), \text{absorption}} \boxed{\frac{SR}{\exists}}
 \end{array}$$

the first three steps only unfold  $\diamond$  —  $E$  is a functor, so  $E\left(\frac{R}{\exists}\right)\text{union} = E\left(\frac{R}{\exists}\exists\right) = E(R)$  and the  $\text{union}$  is gone.

Absorption, the row of (10b), is the whole law, and it is the functoriality of  $\frac{1}{\exists}$ .

$1$  goes to  $\frac{1}{\exists}$ , the Kleisli identity, by definition — so  $\frac{1}{\exists}$  is an isomorphism of categories  $\mathcal{A} \cong \mathbf{Kleisli}(E)$ , and §10.1's  $i \dashv E$  is the Kleisli adjunction.  $i$  is the identity on objects, so every object of the allegory is free.